

Randomized and low-rank approximation

Absolute-error threshold for extremal Gaussian trace bounds

Difficulty: challenging **Importance:** interesting to the community**Status:** Open

Rating rationale: Challenging because signed eigenvalue contributions require uniform extremal tail comparisons; community impact is rigorous trace estimates for indefinite matrices.

Related auxiliary counterexamples — 2026-09-11

Author: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. **Independent Codex-agent review:** PASS for the limited result below.

The centered augmented density has a genuine inflection point beyond the proposed auxiliary upper bound. Together with the mode example, this refutes the upper assertions of Hallman Conjectures 1 and 2.

Remaining question: The complete absolute Gaussian trace-tail probability chain in this entry is neither proved nor refuted. Failure of an auxiliary sufficient condition does not refute the final tail comparisons; status remains Open.

Primary reference: [complete authored PDF](#), [standalone TeX](#), **Proposition 3.1 (with Proposition 2.1 for related evidence)**. [Independent proof review](#) · [Authorship and submission record](#). Verification is independent agent review, not external human peer review or formal certification.

Problem statement

For a real symmetric $d \times d$ matrix D , define

$$T_m(D) = \frac{1}{m} \sum_{j=1}^m z_j^T D z_j, \quad z_1, \dots, z_m \stackrel{\text{iid}}{\sim} N(0, I_d).$$

Let $A \neq 0$ be an arbitrary real symmetric $n \times n$ matrix, possibly indefinite. Put

$$\lambda = \|A\|_2, \quad \phi = \|A\|_F, \quad \rho = \frac{\phi^2}{\lambda^2}, \quad B_{\lambda, \phi} = \lambda \operatorname{diag}(I_{\lfloor \rho \rfloor}, \sqrt{\rho - \lfloor \rho \rfloor}).$$

The norms are the spectral and Frobenius norms. Let X have Gamma shape $m\rho/2$ and rate $m/(2\lambda)$, so that $\mathbb{E}X = \phi^2/\lambda$. Here a Gamma variable with shape α and rate β has density $\beta^\alpha x^{\alpha-1} e^{-\beta x} / \Gamma(\alpha)$ for $x > 0$.

Conjecture. For every integer $n \geq 1$, every such A , every integer $m \geq 1$, and every

$$\varepsilon \geq \frac{2\lambda}{m} + \sqrt{\frac{2\phi^2}{m} + \left(\frac{2\lambda}{m}\right)^2},$$

the following comparisons hold:

$$\begin{aligned} \Pr(|T_m(A) - \operatorname{tr}(A)| \geq \varepsilon) &\leq 2 \Pr(T_m(B_{\lambda, \phi}) - \operatorname{tr}(B_{\lambda, \phi}) \geq \varepsilon) \\ &\leq 2 \Pr\left(X - \frac{\phi^2}{\lambda} \geq \varepsilon\right). \end{aligned}$$

Zero trailing diagonal entries in $B_{\lambda, \phi}$ are harmless. Each probability uses the appropriate estimator dimension. This is an absolute-error question, even when $\operatorname{tr}(A) = 0$.

Why it matters in numerical linear algebra

The bound would quantify trace-estimation error even when positive and negative eigenvalues cancel.

References

1. Eric Hallman, *Extremal bounds for Gaussian trace estimation*, arXiv:2411.15454v1 (2024), §5, Theorem 7 and Conjecture 4.

Status check

Theorem 7 leaves its threshold unspecified. Checked 2026-09-08: the source still lists only v1. Author, title and 2025/2026 searches, including the [later XTrace paper](#), found no resolution. This is a bounded check.

The displayed conjecture is retained despite a source prose inconsistency: its threshold tends to zero as $m \rightarrow \infty$, whereas the following sentence says it tends to ϕ .

Audit — 2026-09-10

Rechecked [Hallman, Theorem 7 and Conjecture 4](#); the record remains v1. Author and extremal-Gaussian-trace searches found no resolution. The displayed formula matches the numbered conjecture; its inconsistent following prose remains disclosed, and no sharper threshold is claimed proved.